

Banking Project:

BASIC + WHERE + OPERATORS :

252

253 -- 1. Retrieve all accounts with balance greater than 20,000.

254

255 • **select** * **from** accounts **where** balance > 20000;

256

257

258

Result Grid | | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	account_id	customer_id	account_type	balance
▶	2	2	Current	32000.50
	3	3	Savings	45500.75
	5	5	Current	87500.00
	6	6	Savings	23450.60
	7	7	Current	99000.25
	9	9	Savings	64000.40
	10	10	Current	28000.90
	12	12	Current	75000.00
	14	14	Savings	36000.00
	15	15	Current	125000.00
	17	17	Current	52000.75
	18	18	Current	10000.00

258

259

260 -- 2. Find customers who live in Chennai.

261

262 • **select** * **from** customers **where** city = 'chennai';

263

264

Result Grid | | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	customer_id	name	city
▶	1	Arun Kumar	Chennai
	5	Karthik Raj	Chennai
	33	Sowmya P	Chennai
	37	Preethi N	Chennai
	45	Shalini Verma	Chennai
	66	Reshma K	Chennai
	88	Tarun Kapoor	Chennai
	100	Balaji K	Chennai
	NULL	NULL	NULL
*			

```

265
266 -- 3. Display accounts with balance between 20,000 and 50,000.
267
268 • select * from accounts where balance between 20000 and 50000;|
269
270

```

Result Grid				
Filter Rows:		Edit:		Export/Import:
				Wrap Cell Content:
account_id	customer_id	account_type	balance	
2	2	Current	32000.50	
3	3	Savings	45500.75	
6	6	Savings	23450.60	
10	10	Current	28000.90	
14	14	Savings	36000.00	
18	18	Savings	43000.25	
19	19	Savings	27800.00	
24	24	Savings	48300.80	
26	26	Savings	26500.00	
27	27	Current	48000.50	
31	31	Savings	30500.80	
33	33	Savings	21250.00	

```

271
272 -- 4. Find customers whose names start with 'M'.
273
274 • select * from customers where name like 'm%';|
275

```

Result Grid			
Filter Rows:		Edit:	
customer_id	name	city	
10	Meena Lakshmi	Salem	
15	Manoj Das	Bhubaneswar	
34	Mahesh Babu	Nellore	
51	Mohan Raj	Chengalpattu	
60	Monika Gupta	Gwalior	
78	Murugan T	Ariyalur	
93	Megha Sharma	Panaji	
NULL	NULL	NULL	

278 -- 5. Retrieve accounts of type 'Savings' or 'Current'.

279

280 • `select * from accounts where account_type = 'savings' or account_type = 'current';`

281

282

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	account_id	customer_id	account_type	balance
▶	1	1	Savings	15000.00
	2	2	Current	32000.50
	3	3	Savings	45500.75
	4	4	Savings	12000.00
	5	5	Current	87500.00
	6	6	Savings	23450.60
	7	7	Current	99000.25
	8	8	Savings	17890.00
	9	9	Savings	64000.40
	10	10	Current	28000.90

283

284 -- 6. Display accounts that are not 'Savings'.

285

286 • `select * from accounts where account_type != 'Savings';`

287

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	account_id	customer_id	account_type	balance
▶	2	2	Current	32000.50
	5	5	Current	87500.00
	7	7	Current	99000.25
	10	10	Current	28000.90
	12	12	Current	75000.00
	15	15	Current	125000.00
	17	17	Current	52000.75
	20	20	Current	88000.10
	22	22	Current	67000.65
	25	25	Current	102500.00
	27	27	Current	48000.50
	30	30	Current	84500.25

289

290 -- 7. Find customers whose names contain the letter 'a'.

291

292 • **select** * **from** customers **where** name like '%a%';

293

Result Grid   Filter Rows: Edit:    Export/Import:   Wrap Cell Content: 			
	customer_id	name	city
▶	1	Arun Kumar	Chennai
	2	Priya Sharma	Bengaluru
	3	Rahul Verma	Hyderabad
	4	Sneha Reddy	Mumbai
	5	Karthik Raj	Chennai
	6	Divya Nair	Kochi
	7	Vikram Singh	Delhi
	8	Anjali Gupta	Pune
	9	Suresh Babu	Madurai
	10	Meena Lakshmi	Salem

295

296 -- 8. Retrieve accounts with balance less than or equal to 30,000.

297

298 • **select** * **from** accounts **where** balance <= 30000;

299

Result Grid   Filter Rows: Edit:    Export/Import:   Wrap Cell Content: 				
	account_id	customer_id	account_type	balance
▶	1	1	Savings	15000.00
	4	4	Savings	12000.00
	6	6	Savings	23450.60
	8	8	Savings	17890.00
	10	10	Current	28000.90
	11	11	Savings	5400.75
	13	13	Savings	18400.20
	16	16	Savings	9600.50
	19	19	Savings	27800.00
	21	21	Savings	14000.00
	23	23	Savings	9200.30

301

302 -- 9. Find customers who are not from Chennai.

303

304 • **select * from** customers **where** city **not in** ('Chennai');

305

Result Grid			
Filter Rows:			
Edit:   			
Export/Import:  			
Wrap Cell Content: 			
	customer_id	name	city
▶	2	Priya Sharma	Bengaluru
	3	Rahul Verma	Hyderabad
	4	Sneha Reddy	Mumbai
	6	Divya Nair	Kochi
	7	Vikram Singh	Delhi
	8	Anjali Gupta	Pune
	9	Suresh Babu	Madurai
	10	Meena Lakshmi	Salem
	11	Ramesh Kumar	Trichy
	12	Pooja Mehta	Ahmedabad
	13	Ajith Kumar	Vellore

307

308 -- 10. Display accounts where balance is not between 10,000 and 40,000.

309

310 • **select * from** accounts **where** balance **not between** 10000 and 40000;

311

Result Grid				
Filter Rows:				
Edit:   				
Export/Import:  				
Wrap Cell Content: 				
	account_id	customer_id	account_type	balance
▶	3	3	Savings	45500.75
	5	5	Current	87500.00
	7	7	Current	99000.25
	9	9	Savings	64000.40
	11	11	Savings	5400.75
	12	12	Current	75000.00
	15	15	Current	125000.00
	16	16	Savings	9600.50
	17	17	Current	52000.75
	18	18	Savings	43000.25
	20	20	Current	88000.10

313

314 -- 11. Retrieve customers whose names end with 'i'.

315

316 • **select** * **from** customers **where** name **like** '%i';

317

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:			
	customer_id	name	city
▶	10	Meena Lakshmi	Salem
	20	Asha Devi	Visakhapatnam
	35	Geetha Lakshmi	Kurnool
	39	Bhavani Devi	Udupi
	44	Amit Tiwari	Lucknow
	48	Sunil Joshi	Dehradun
	49	Pallavi Kulkarni	Nashik
	52	Kavya Sri	Dindigul

319

320 -- 12. Find accounts with balance equal to 75,000.

321

322 • **select** * **from** accounts **where** balance = 75000;

323

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:				
	account_id	customer_id	account_type	balance
▶	12	12	Current	75000.00
*	NULL	NULL	NULL	NULL

326 -- 13. Display customers whose city is either Chennai or Salem.

327

328 • **select** * **from** customers **where** city **in** ('Chennai', 'Salem');

329

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:			
	customer_id	name	city
▶	1	Arun Kumar	Chennai
	5	Karthik Raj	Chennai
	10	Meena Lakshmi	Salem
	33	Sowmya P	Chennai


```

331
332  -- 14. Find accounts with balance greater than 10,000 and less than 40,000.
333
334 • select * from accounts where balance > 10000 and balance < 40000;
335

```

Result Grid				
Filter Rows:				
Edit:   				
Export/Import:  				
Wrap Cell Content: 				
	account_id	customer_id	account_type	balance
▶	1	1	Savings	15000.00
	2	2	Current	32000.50
	4	4	Savings	12000.00
	6	6	Savings	23450.60
	8	8	Savings	17890.00
	10	10	Current	28000.90
	13	13	Savings	18400.20
	14	14	Savings	36000.00
	19	19	Savings	27800.00

```

337
338  -- 15. Retrieve accounts where account type is not in ('Current').
339
340 • select * from accounts where account_type not in ('Current');
341

```

Result Grid				
Filter Rows:				
Edit:   				
Export/Import:  				
Wrap Cell Content: 				
	account_id	customer_id	account_type	balance
▶	1	1	Savings	15000.00
	3	3	Savings	45500.75
	4	4	Savings	12000.00
	6	6	Savings	23450.60
	8	8	Savings	17890.00
	9	9	Savings	64000.40
	11	11	Savings	5400.75
	13	13	Savings	18400.20
	14	14	Savings	36000.00
	16	16	Savings	9600.50

Order by :

```

343
344 -- 16. Display all accounts sorted by balance in descending order.
345
346 • select * from accounts order by balance desc;
347

```

Result Grid   Filter Rows: | Edit:    | Export/Import:   | Wrap Cell Content: 

	account_id	customer_id	account_type	balance
▶	100	100	Current	150250.00
	50	50	Current	150000.00
	75	75	Current	145800.00
	64	64	Current	137500.25
	86	86	Current	133800.00
	37	37	Current	132000.00
	92	92	Current	129400.50
	71	71	Current	126000.00

```

349
350 -- 17. List customers sorted alphabetically by name.
351
352 • select * from customers order by name asc;
353

```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	customer_id	name	city
	32	Arvind Rao	Warangal
	20	Asha Devi	Visakhapatn...
	38	Ashok Kumar	Shimoga
	100	Balaji K	Chennai
	28	Bharath Kumar	Hosur
	39	Bhavani Devi	Udupi
	22	Deepika Rao	Mangalore
	67	Dinesh Kumar	Tiruvannam...
	95	Divakar R	Puducherry
	6	Divya Nair	Kochi


```

355
356 -- 18. Display accounts sorted by account type and then by balance (descending).
357
358 • select * from accounts order by account_type desc, balance desc;
359

```

account_id	customer_id	account_type	balance
16	16	Savings	9600.50
23	23	Savings	9200.30
11	11	Savings	5400.75
100	100	Current	150250.00
50	50	Current	150000.00
75	75	Current	145800.00
64	64	Current	137500.25
86	86	Current	133800.00

Aggregate Functions :

```

361
362 -- 19. Find the total balance of all accounts.
363
364 • select sum(balance) as 'Total_Balance' from accounts;
365

```

Total_Balance
5646772.70

```

367
368 -- 20. Calculate the average balance of accounts.
369
370 • select avg(balance) as 'Average_Balance' from accounts;
371

```

Average_Balance
56467.727000

```
373
374 -- 21. Find the maximum account balance.
375
376 • select max(balance) as 'Maximum_Balance' from accounts;
377
```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	Maximum_Balance
▶	150250.00

```
379
380 -- 22. Find the minimum account balance.
381
382 • select min(balance) 'Minimum_Balance' from accounts;
383
```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	Minimum_Balance
▶	5400.75

```
385
386 -- 23. Count the total number of customers.
387
388 • select count(*) 'Total_Customers' from customers;
389
```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	Total_Customers
▶	100

Group by + Having :

```
391
392 -- 24. Find total balance grouped by account type.
393
394 • select account_type, sum(balance) 'total_balance' from accounts group by account_type;
395
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
account_type	total_balance		
▶ Savings	1665208.45		
Current	3981564.25		

```
398 -- 25. Find average balance for each account type.
399
400 • select account_type, avg(balance) 'average_balance' from accounts group by account_type;
401
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
account_type	average_balance		
▶ Savings	29214.183333		
Current	92594.517442		

```
404 -- 26. Display account types having average balance greater than 20,000.
405
406 • select account_type, avg(balance) 'average_balance' from accounts group by account_type having avg(balance) > 20000;
407
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
account_type	average_balance		
▶ Savings	29214.183333		
Current	92594.517442		

```
409
410 -- 27. Count number of accounts for each customer.
411
412 • select customer_id, count(account_id) 'total_accounts' from accounts group by customer_id;
413
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
customer_id	total_accounts		
▶ 1	6		
2	1		
3	5		
4	1		
6	4		
7	1		
9	1		
10	3		


```

416 -- 28. Display customers having more than one account.
417
418 • select customer_id, count(account_id) total_accounts from accounts
419    group by customer_id having count(account_id) > 1;
420

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
customer_id	total_accounts			
1	6			
3	5			
6	4			
10	3			

Joins :

```

421
422 -- 29. Retrieve customer names along with their account balances.
423
424 • select c.name, a.balance from customers c join accounts a on c.customer_id = a.customer_id;
425

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
name	balance			
Meena Laks...	48000.50			
Meena Laks...	62500.50			
Ramesh Ku...	5400.75			
Pooja Mehta	75000.00			
Ajith Kumar	18400.20			
Neha Agarwal	36000.00			
Manoj Das	125000.00			
Lakshmi Priya	9600.50			

```

427 -- 30. Display all customers and their accounts (including customers without accounts).
428
429 • select c.customer_id, c.name, c.city, a.account_id, a.account_type, a.balance
430    from customers c left join accounts a on c.customer_id = a.customer_id;
431

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

customer_id	name	city	account_id	account_type	balance
6	Divya Nair	Kochi	50	Current	150000.00
6	Divya Nair	Kochi	60	Savings	33800.30
7	Vikram Singh	Delhi	7	Current	99000.25
8	Anjali Gupta	Pune	NULL	NULL	NULL
9	Suresh Babu	Madurai	9	Savings	64000.40
10	Meena Laks...	Salem	10	Current	28000.90

432

433 -- 31. Display all accounts and corresponding customer details.

434

435 • `select c.customer_id, c.name, c.city, a.account_id, a.account_type, a.balance`

436 `from customers c inner join accounts a on c.customer_id = a.customer_id;`

437

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

customer_id	name	city	account_id	account_type	balance
66	Reshma K	Chennai	66	Current	68800.10
68	Vaishnavi R	Pudukkottai	68	Savings	25400.45
69	Lokesh Kumar	Ramanat...	69	Current	83200.90
70	Anitha S	Nagercoil	70	Savings	44750.00
71	Nitin Agarwal	Agra	71	Current	126000.00
72	Pavithra N	Karaikudi	72	Savings	16200.35
73	Srinivas Rao	Khammam	73	Current	59800.70
74	Harini V	Ooty	74	Savings	27100.25
75	Vigneshwar...	Kallakurichi	75	Current	145800.00

438

439 -- 32. Retrieve customer names and account types where balance is greater than 20,000.

440

441 • `select c.name, a.account_type from customers c join accounts a`

442 `on c.customer_id = a.customer_id where a.balance > 20000;`

443

Result Grid

 Filter Rows:

Export: 

Wrap Cell Content: 

	name	account_type
▶	Priya Sharma	Current
	Rahul Verma	Savings
	Arun Kumar	Current
	Divya Nair	Savings
	Vikram Singh	Current
	Suresh Babu	Savings
	Meena Lakshmi	Current
	Pooja Mehta	Current

445 -- 33. List customers with their total balance using JOIN.

446

447 • `select c.name, sum(a.balance) as total_balance from customers c join accounts a`

448 `on a.customer_id = c.customer_id group by c.name;`

449

Result Grid





Filter Rows:

Export:



Wrap Cell Content:



	name	total_balance
▶	Arun Kumar	259152.00
	Priya Sharma	32000.50
	Rahul Verma	175193.05
	Sneha Reddy	12000.00
	Divya Nair	291751.15
	Vikram Singh	99000.25

```

450
451 -- 34. Display customer names and balances sorted by balance.
452
453 • select c.name, a.balance from customers c join accounts a
454   on a.customer_id = c.customer_id order by balance asc;
455

```

Result Grid		
Filter Rows:		
Export:  Wrap Cell Content: 		
	name	balance
▶	Ramesh Kumar	5400.75
	Ganesh Kumar	9200.30
	Lakshmi Priya	9600.50
	Arun Kumar	9800.00
	Vinod Kumar	11800.20
	Sneha Reddy	12000.00
	Anu Priya	13500.00
	Rohit Sharma	14000.00
	Gayathri R	14200.00
	Arun Kumar	15000.00
	Lavanya Devi	15250.75
	Megha Sharma	15600.40
	Nandhini V	15900.00

```

457 -- 35. Count number of accounts for each city using JOIN.
458
459 • select c.city, count(a.account_id) as total_city_accounts from customers c join accounts a
460   on c.customer_id = a.customer_id group by c.city;
461

```

Result Grid		
Filter Rows:		
Export:  Wrap Cell Content: 		
	city	total_city_accounts
▶	Chennai	11
	Bengaluru	1
	Hyderabad	5
	Mumbai	1
	Kochi	4
	Delhi	1
	Madurai	1
	Salem	3
	Trichy	1
	Ahmedabad	1
	Vellore	1

Sub-queries :

463 -- 36. Find accounts with balance greater than average balance.

464

465 • `select * from accounts where balance > (select avg(balance) from accounts);`

466

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:				
	account_id	customer_id	account_type	balance
▶	5	1	Current	87500.00
	7	7	Current	99000.25
	9	9	Savings	64000.40
	12	12	Current	75000.00
	15	15	Current	125000.00
	20	20	Current	88000.10
	22	22	Current	67000.65
	25	25	Current	102500.00

468 -- 37. Retrieve customers who have accounts.

469

470 • `select * from customers where customer_id in (select customer_id from accounts);`

471

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:			
	customer_id	name	city
▶	1	Arun Kumar	Chennai
	2	Priya Sharma	Bengaluru
	3	Rahul Verma	Hyderabad
	4	Sneha Reddy	Mumbai
	6	Divya Nair	Kochi
	7	Vikram Singh	Delhi
	9	Suresh Babu	Madurai
	10	Meena Lakshmi	Salem
	11	Ramesh Kumar	Trichy
	12	Pooja Mehta	Ahmedabad

474 -- 38. Find customers who do not have any accounts.

475

476 • `select * from customers where customer_id not in (select customer_id from accounts);`

477

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:			
	customer_id	name	city
▶	5	Karthik Raj	Chennai
	8	Anjali Gupta	Pune
	17	Harish Kumar	Mysuru
	27	Kavitha R	Namakkal

479 -- 39. Display account(s) with the maximum balance.

480

481 • `select * from accounts where balance = ( select max(balance) from accounts);`

482

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
account_id	customer_id	account_type	balance	
100	100	Current	150250.00	
NULL	NULL	NULL	NULL	

483

484 -- 40. Find customers whose total balance is greater than 40,000.

485

486 • `select * from customers where customer_id in (select customer_id from accounts where balance > 40000);`

487

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
customer_id	name	city		
3	Rahul Verma	Hyderabad		
1	Arun Kumar	Chennai		
7	Vikram Singh	Delhi		
9	Suresh Babu	Madurai		
12	Pooja Mehta	Ahmedabad		
15	Manoj Das	Bhubaneswar		
18	Keerthana S	Erode		
20	Asha Devi	Visakhapatnam		
22	Deepika Rao	Mangalore		
24	Swetha R	Tiruppur		
25	Prakash R	Dharmapuri		
10	Meena Laks...	Salem		
28	Bharath Ku...	Hosur		
6	Divya Nair	Kochi		